

Lecture 8: Estimation

Computational Bootcamp

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Today's goals

1. Estimation is just optimization
2. GMM: moment conditions, the weighting matrix, and specification test
3. Know what changes when the moments have to be simulated

Estimation as Optimization

- An estimator returns $\hat{\theta}$, defined as *“the θ that minimizes something”*.

	minimize	how
OLS	sum of squared residuals	closed form
MLE	$-\log L(\theta)$	Optim
GMM	$\bar{g}(\theta)' W \bar{g}(\theta)$	Optim
SMM	the same, with $\bar{g}$ simulated	Optim + shock draws

- In structural work, evaluating the objective means solving the model: an optimizer over θ outside, VFI or Huggett economy inside

- **CSV** + **DataFrames**: read the file, hold the table. Columns by name, select/filter, groupby + combine, joins — and `Matrix(df[:, [:educ, :exp]])` when an estimator wants an array
- **Statistics** is a stdlib, nothing to install: mean, std, cor, quantile — your data moments
- **StatsModels**' `@formula` builds the design matrix — interactions and categorical dummies. **GLM** fits it: `lm/glm`, then `coef`, `stderror`, `coeftable`

```
using CSV, DataFrames, Statistics, GLM      # GLM re-exports @formula
df = CSV.read("data/lwage.csv", DataFrame)
df.exp2 = df.exp .^ 2
m = lm(@formula(lwage ~ educ + exp + exp2), df) # OLS, s.e.s, t-stats
mean(df.lwage), cor(df.educ, df.lwage)         # moments to match later
```

- **GLM** fits models someone else wrote. You should be able to write it yourself

Maximum Likelihood

Maximum likelihood

- **The idea:** pick parameter value that makes the data you observe most probable. Logs because products of n small numbers underflow, and because a sum has a derivative

$$\log L(\theta) = \sum_{i=1}^n \log f(y_i | x_i, \theta)$$

- Hand Optim $-\log L$, and remember the sign when you read the output
- **Parameters have constraints; most optimizers don't.** $\sigma > 0$, but a line search will try $\sigma = -3$. Fix it with a change of variables or a penalty.

```
function neg_loglik(theta, X, y)
  b, s = theta[1:end-1], exp(theta[end])  # <- s > 0 for free
  e, n = y .- X*b, length(y)
  return n*log(s) + n*log(2pi)/2 + sum(abs2, e)/(2*s^2)
end
```

- Trick: stitching functions - use a linear approximation of the objective close to the constraint to preserve continuity and gradient

Standard errors from curvature

- A likelihood sharply peaked at $\hat{\theta}$ says the data are informative; a flat one says they are not.

Curvature is information:

$$\hat{V}(\hat{\theta}) = \left[\underbrace{\nabla^2 \left(-\log L(\hat{\theta}) \right)}_{\text{observed information}} \right]^{-1}, \quad \text{s.e.} = \sqrt{\text{diag } \hat{V}}$$

- Hessian by automatic differentiation, or by central differences in four lines. The bootstrap is the other route: no derivatives, many more model solves
- Optimized over $\log \sigma$? The Hessian gives you $\text{s.e.}(\widehat{\log \sigma})$. The **delta method** converts it
- **Diagnostics**: did the optimizer converge? Is the Hessian positive definite?

Likelihoods in economics

The density $f(y_i | x_i, \theta)$ is rarely handed to you — it is an output of the model.

- **Discrete choice.** The “density” is a choice probability. Logit gives it to you in closed form; probit and mixed logit make it an integral over tastes, so you *simulate* it (simulated MLE)
- **Dynamic discrete choice** (Rust 1987, bus engines). The probability a manager replaces an engine comes out of a Bellman equation. Evaluating the likelihood at one θ means **solving the dynamic program at that θ** — nested fixed point algorithm
 - Outer loop: Optim over θ . Inner loop: VFI.
- **Demand estimation.** Micro data on who bought what gives you a likelihood

The pattern to take away: the econometrics is a wrapper. The expensive, breakable part is the model solve inside it.

GMM

Generalized method of moments

- MLE needs a distribution. Often your theory does not give you that — it gives you moments or things that are zero in expectation.
- A **moment condition** is exactly that: $\mathbb{E}[g(w_i, \theta_0)] = 0$, true at the true parameter
 - Exogeneity / IV: $\mathbb{E}[z_i(y_i - x_i'\beta)] = 0$
 - An Euler equation (Hansen–Singleton): $\mathbb{E}\left[\beta(1 + r_{t+1})\frac{u'(c_{t+1})}{u'(c_t)} - 1 \mid \mathcal{I}_t\right] = 0$ — **no distributional assumption anywhere**
- Replace the expectation with its sample analogue $\bar{g}(\theta) = \frac{1}{n} \sum_i g(w_i, \theta)$:

$$\hat{\theta} = \arg \min_{\theta} \bar{g}(\theta)' W \bar{g}(\theta), \quad W \text{ positive definite}$$

- **This is the workhorse.** Almost every structural estimator you will meet — SMM, indirect inference, BLP — is this with a different g

The weighting matrix, and what over-identification buys

Let m = number of moments, k = number of parameters.

- $m = k$ (just-identified). You can usually drive $\bar{g}(\hat{\theta})$ to *exactly* zero, so W has nothing to trade off.
- $m > k$ (over-identified). The moments may not all hold at once. W decides which ones give, so W changes your estimate
- Efficient choice: $W = S^{-1}$ with $S = \text{Var}(\sqrt{n} \bar{g})$ — down-weight the moments your data measure noisily. But S needs a $\hat{\theta}$ and $\hat{\theta}$ needs W , hence two-step GMM

Simulated Moments

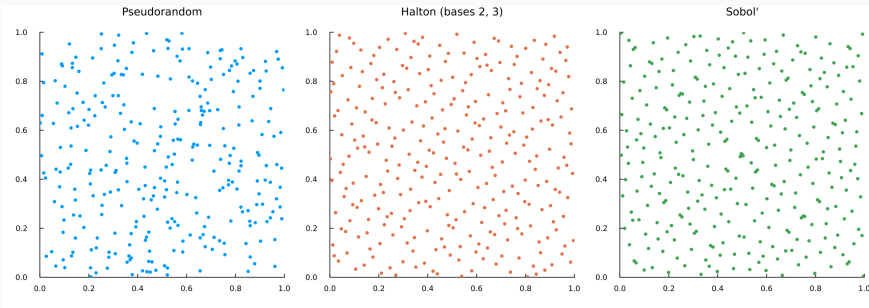
When the moment has no formula

- Often you cannot compute $g(w_i, \theta)$ by hand: the model moment is the output of a solved model — mixed logit shares, or the wealth distribution in a Huggett economy
- **Simulate it.** Draw shocks once, build a synthetic dataset at the candidate θ , compute the moments on it, then minimize via GMM (or SMM)

$$(m^{\text{data}} - m^{\text{sim}}(\theta))' W (m^{\text{data}} - m^{\text{sim}}(\theta))$$

- **Draw the shocks ONCE, outside the objective, and pass them in.** Redraw inside and the optimizer reads sampling noise between two nearby points as a gradient
- Average the *moments* across draws, not the draws across moments — these differ for anything nonlinear, and variances and shares are nonlinear

A simulated moment is an integral, so integrate well



- Random points clump and leave holes, so error falls only like $1/\sqrt{R}$. Halton and Sobol' points fill the cube by construction — roughly an order of magnitude at the same R
- Bhat (2001): 100 Halton draws beat 1000 random draws in a mixed logit — standard practice in demand estimation. Sobol works better in high dimensions, can be scrambled

Where To Go Next

The on-ramp to Econ 880

- Econ 880 (Computational Economics):
 - Heterogeneous agent models, firm dynamics
 - Estimation via SMM / indirect inference
 - Extensions (aggregate risk)
- QuantEcon (<https://julia.quantecon.org/>); Fernández-Villaverde's notes (<https://www.sas.upenn.edu/~jesusfv/>)
- Pick an economic question *you* care about, write the model down, build it. Learn by doing.
- Let me know if you have any questions. I was the “TA” for Econ 880

To the code, one last time!